

Advanced (Habanero)

Q1. Factor the expression $x^2 - 4$

- (A) $(x + 2)(x - 2)$
- (B) $x(x - 4)$
- (C) $(x + 2)^2$
- (D) $(x - 2)^2$
- (E) $x^2 - 2x + 4$

Q2. Which of the following is a factored form of $49x^2 - 9$?

- (A) $(7x + 3)(7x - 3)$
- (B) $(7x - 3)(7x + 3)$
- (C) $(3x + 7)(3x - 7)$
- (D) $9(7x^2 - 1)$
- (E) $49(x^2 - 1)$

Q3. Factor $36a^2 - 24b^2$

- (A) $12(3a - 2b)(3a + 2b)$
- (B) $12(3a + 2b)(3a - 2b)$
- (C) $12(2a - 3b)(2a + 3b)$
- (D) $6(6a^2 - 4b^2)$
- (E) $12(6a^2 - 2b^2)$

Q4. What is the factored form of $16y^2 - 25z^2$?

- (A) $(4y + 5z)(4y - 5z)$
- (B) $(5z + 4y)(5z - 4y)$
- (C) $(4y - 5z)(4y + 5z)$
- (D) $(16y^2 - 25z^2)$
- (E) $(4y - 5z)(4y - 5z)$

Q5. Factor $81x^2 - 16y^2$

- (A) $(9x + 4y)(9x - 4y)$
- (B) $(9x - 4y)(9x + 4y)$
- (C) $(16y^2 - 81x^2)$
- (D) $9(9x^2 - 16y^2)$
- (E) $16(16y^2 - 81x^2)$

Q6. Factor $100a^2 - 9b^2$

- (A) $10(10a + 3b)(10a - 3b)$
- (B) $10(10a - 3b)(10a + 3b)$
- (C) $9(9a^2 - 100b^2)$
- (D) $(10a - 3b)(10a + 3b)$
- (E) $(100a^2 - 9b^2)$

Q7. Which is the factored form of $64x^2 - 121y^2$?

- (A) $(8x + 11y)(8x - 11y)$
- (B) $(8x - 11y)(8x + 11y)$
- (C) $(64x^2 - 121y^2)$
- (D) $64(x^2 - y^2)$
- (E) $121(y^2 - x^2)$

Q8. What is the factored form of $400a^2 - 169b^2$?

- (A) $20(20a - 169b)(20a + 169b)$
- (B) $20(20a + 169b)(20a - 169b)$
- (C) $(20a + 13b)(20a - 13b)$
- (D) $169(169b^2 - 400a^2)$
- (E) $(20a - 13b)(20a + 13b)$

Open Ended Questions (FRQ)

Q9. Factor the expression $x^4 - y^4$. Use the difference of squares pattern and show all the steps of your work.

Q10. Factor $196x^2 - 36y^2$ completely. Justify each step with the pattern being used.

Chef's Tips

- Make sure students show all the steps of their work on free response questions.
- Emphasize that students should check their work by expanding the factored form.
- Provide extra help to students who are having trouble applying the difference of squares pattern.